

ERBOT Results Summary

NCVR 5-Party Voter Registration Dataset

ERBOT v0.2.0

Generated: 2026-03-25 | Small run: <1 hr | Large run: $\approx$ 45 hrs

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1 Dataset Overview

Property	Value
Dataset	NCVR 5-Party Voter Registration (ocp=20, modrec=2)
Source files	ncvr_numrec_1000000_modrec_2_ocp_20_myp_{0,1}_nump_5.csv
Parties used	2 (party 0 and party 1 from the 5-party collection)
Fields	4 content fields: <code>givenname</code> , <code>surname</code> , <code>suburb</code> , <code>postcode</code>
Task	Record linkage (two-source)
Overlap rate	20% (ocp=20); modification rate: 2 fields per duplicate (modrec=2)
True matches	$\approx$ 880 cross-party pairs (varies slightly by sample)

The NCVR 5-Party benchmark is derived from the North Carolina Voter Registration database. The generator produces N records per party with a controlled overlap percentage and modification rate. Records that appear in both parties share the same `recid` but may differ in up to `modrec`

fields. This benchmark tests cross-party record linkage under realistic personal-name and address noise. Two runs were conducted: a **small run** (5K records per party) and a **large run** (50K records per party) to assess scalability.

2 Pipeline Configuration

Stage	Setting	Small (5K/party)	Large (50K/party)
Input	Records total	10,000	100,000
Input	True matches	≈ 880	≈ 880
Input	Match density	$\sim 8.8\%$	$\sim 0.88\%$
Blocking	Method	Prefix-3 on surname	Prefix-3 on surname
Blocking	Candidate pairs	$\sim 10,600$ (est.)	1,059,972 (truncated from 2,121,188)
Blocking	Reduction ratio	≈ 1.000	1.000
Similarity	Fields computed	4	4
Weights	Method	Equal (0.25 each)	Equal (0.25 each)
Clustering	Methods run	All 9 (incl. SVM)	8 (SVM excluded; hclust/pam/gc failed)
Merge	Strategy	Best	Best
Final	Clusters	1,910	94,706

Blocking note: Prefix-3 on **surname** achieves near-total pair reduction ($RR \approx 1.000$ due to rounding at this scale). In the large run the blocker generated 2,121,188 raw pairs which exceeded the 2M pair budget and was truncated to approximately 2M; the final sparse matrix has 2,119,944 non-zero entries. This truncation may silently drop true match pairs and directly suppresses recall.

Match density note: Both runs target the same absolute number of true matches (≈ 880) because the ground truth is defined by recid overlap in the sampled subsets. The large run has $10\times$ more records but the same matches, giving a $10\times$ lower match density (0.88% vs. 8.8%). This is the primary driver of the performance gap between small and large runs.

3 Performance Results — Small Run (5 000 records/party)

Method	ARI	NMI	VI	B ³ -P	B ³ -R	B ³ -F	Hom.	Com.	V	PF-P	PF-R	PF-F
threshold_cc	0.917	0.975	0.143	1.000	0.929	0.963	1.000	0.952	0.975	1.000	0.857	0.923
louvain	0.917	0.975	0.143	1.000	0.929	0.963	1.000	0.952	0.975	1.000	0.857	0.923
label_prop	0.917	0.975	0.143	1.000	0.929	0.963	1.000	0.952	0.975	1.000	0.857	0.923
svm	0.917	0.975	0.143	1.000	0.929	0.963	1.000	0.952	0.975	1.000	0.857	0.923
hclust_avg	0.276	0.713	1.315	0.537	0.929	0.681	0.583	0.920	0.713	0.231	0.857	0.364
pam	0.158	0.672	1.601	0.537	0.786	0.638	0.583	0.792	0.672	0.167	0.571	0.258
leiden	0.000	0.849	1.000	1.000	0.500	0.667	1.000	0.737	0.849	0.000	0.000	0.000
hclust_ward	0.068	0.542	1.944	0.429	0.857	0.571	0.409	0.801	0.542	0.111	0.714	0.192
gc	0.000	0.000	2.807	0.143	1.000	0.250	0.000	1.000	0.000	0.077	1.000	0.143
<i>final</i>	0.917	0.975	0.143	1.000	0.929	0.963	1.000	0.952	0.975	1.000	0.857	0.923

4 Performance Results — Large Run (50 000 records/party)

* Method failed at runtime and fell back to degenerate singleton assignment. *hclust_avg/ward* and *pam* exceeded *R*'s 65,536-observation distance-matrix limit; *gc* attempted a 74.5 GiB dense allocation. These methods are now excluded from *run_ncvr5_large.R*.

Method	ARI	NMI	VI	B ³ -P	B ³ -R	B ³ -F	Hom.	Com.	V	PF-P	PF-R	PF-F
threshold_cc	0.195	0.956	0.908	0.991	0.560	0.715	0.997	0.917	0.956	0.533	0.119	0.195
louvain	0.133	0.947	1.073	0.933	0.560	0.700	0.980	0.916	0.947	0.152	0.119	0.134
label_prop	0.133	0.947	1.073	0.933	0.560	0.700	0.980	0.916	0.947	0.152	0.119	0.134
leiden	0.000	0.951	1.000	1.000	0.500	0.667	1.000	0.907	0.951	0.000	0.000	0.000
hclust_avg*	0.000	0.000	9.781	0.001	1.000	0.002	0.000	1.000	0.000	0.001	1.000	0.001
hclust_ward*	0.000	0.000	9.781	0.001	1.000	0.002	0.000	1.000	0.000	0.001	1.000	0.001
pam*	0.000	0.000	9.781	0.001	1.000	0.002	0.000	1.000	0.000	0.001	1.000	0.001
gc*	0.000	0.000	9.781	0.001	1.000	0.002	0.000	1.000	0.000	0.001	1.000	0.001
<i>final</i>	0.195	0.956	0.908	0.991	0.560	0.715	0.997	0.917	0.956	0.533	0.119	0.195

Metrics: ARI = Adjusted Rand Index; NMI = Normalized Mutual Information; VI = Variation of Information (lower is better); B³ = B-cubed; V = V-measure; PF = Pairwise F-score; Hom. = Homogeneity; Com. = Completeness.

5 Scale Comparison

Metric	Small (5K/party)	Large (50K/party)	Comment
Records	10,000	100,000	10× scale-up
True matches	~880	~880	Same absolute c
Match density	8.8%	0.88%	10× sparser
Best ARI	0.917	0.195	−0.72 drop
Best B ³ -F	0.963	0.715	−0.25 drop
Best method	threshold_cc / louvain / label_prop / svm (tied)		threshold_cc Consistent winn
Final correct	Yes	Yes	Merge logic worl
Final clusters	1,910	94,706	Most records in
Blocking RR	≈1.000	1.000	Identical (round
B ³ Precision	1.000	0.991	Maintained well
B ³ Recall	0.929	0.560	Major recall dro
Runtime	<1 hr	≈45 hrs	hclust/pam/gc f

The performance collapse from small to large is driven almost entirely by **recall**: precision stays high (1.0 → 0.99) but recall drops from 0.93 to 0.56. At 50K records per party, the sparse surname prefix blocks contain so many non-match pairs that true match pairs are diluted, and some are truncated by the pair budget cap. This is a blocking problem, not a clustering problem.

6 Final Cluster Assignment

In both runs the **final** merge strategy correctly selected **threshold_cc** (highest ARI).

Statistic	Small	Large
Total clusters	1,910	94,706
Records	10,000	100,000
Mean cluster size	5.2	1.06

The large run’s 94,706 clusters from 100,000 records (mean size ≈1.06) indicates that nearly all records are assigned as singletons — correctly reflecting the low match density (only 880 true matches exist in 100K records). The small run’s 1,910 clusters from 10,000 records (mean 5.2) is more compact but reflects the much higher match density at that scale.

7 Method-by-Method Interpretation

Threshold-CC (Best overall — ARI 0.917 / 0.195)

The consistent winner at both scales. At small scale it ties with louvain, label_prop, and svm at ARI = 0.917 and perfect precision ($B^3\text{-P} = 1.0$). At large scale it retains near-perfect precision (0.991) but recall falls to 0.560. The connected-components approach is well-suited to the NCVR similarity graph: at a threshold of 0.5 the graph separates cleanly, and the four-field equal-weight profile creates a well-calibrated similarity score for voter records with controlled 2-field modifications.

Louvain / Label Propagation (Tied-best small; second large)

Identical to threshold-CC at small scale; slightly weaker at large scale (ARI 0.133 vs. 0.195). This divergence suggests that at low match density the community structure in the graph is less clean, and Louvain/LP begin to over-merge some blocks. The lower PairF-P (0.152 vs. 0.533 for threshold-CC) at large scale confirms more false-positive links.

SVM (Best small; not run large)

Ties at ARI = 0.917 in the small run. SVM is excluded from the large run script (correctly — it is too slow at 100K scale). At 5K/party scale it adds no benefit over the threshold-CC family on this structured dataset, but it would be worth including in any small-scale parameter sweep.

Leiden (Degenerate at both scales — ARI = 0.000)

Perfect precision (1.0) and completeness (0.74/0.91) but ARI = 0 and PairF = 0 at both scales. Leiden over-partitions: it splits true duplicate pairs into separate singleton clusters rather than merging them. The resolution parameter is too high for NCVR’s similarity graph topology. Needs tuning.

hclust_avg (Partial failure)

Works at small scale (ARI = 0.276, second-best after the graph quartet) but fails at large scale with R’s 65,536 observation limit. Even at small scale it underperforms graph methods substantially.

hclust_ward / PAM / GC (Fail or degenerate)

hclust_ward and PAM fail at large scale (65K limit). GC degenerates at both scales (ARI = 0): at small scale it produces a near-singleton partition with perfect recall ($B^3\text{-R} = 1.0$) but near-zero precision (0.143); at large scale it crashes with a 74.5 GiB allocation attempt. GC is not viable for NCVR.

8 Key Findings

1. **Threshold-CC is the best and most scalable method.** It wins at both scales, is fast, and avoids the memory failures that affect distance-matrix methods. It is the correct default for NCVR-style 4-field voter linkage.
2. **The performance drop from small to large is a blocking/density problem, not a clustering problem.** Precision is maintained (0.991) but recall drops from 0.929 to 0.560. At 50K/party, true match pairs are sparser and some are truncated by the 2M pair budget cap. The clustering algorithm cannot recover pairs that blocking discards.

3. **Pair budget truncation is active at large scale.** The blocker found 2,121,188 raw pairs ($>2\text{M}$ budget) and truncated to $\approx 2\text{M}$. Some true match pairs may have been silently dropped at this step. The budget should be raised or a smarter blocking strategy adopted.
4. **hclust_avg/ward, PAM, and GC are not viable at 100K scale.** All three fail with R’s 65K distance-matrix limit or memory exhaustion. They have been removed from the large run script. They should also be flagged as “small-scale only” in documentation.
5. **Equal weights work well for NCVR.** With 4 equally noisy fields (name, surname, suburb, postcode), equal weighting is a reasonable choice. Unlike CORA (15 fields of varying quality), there is no evidence that weight learning would substantially improve results here.
6. **The final merge strategy is working correctly.** It selected threshold-CC (highest ARI) in both runs. The earlier merge bug seen in CORA has been resolved.
7. **Leiden consistently degenerates.** $\text{ARI} = 0$ at both scales across NCVR and CORA suggests the default resolution parameter is unsuitable for these datasets. Leiden should not be included in production runs without prior resolution tuning.

9 Recommendations for Next Steps

Priority	Action
High	Increase pair budget beyond 2M (or make it adaptive) to avoid silent truncation of true matches at large scale.
High	Add pair completeness measurement at Stage 3 to quantify how many true matches the blocker discards. Without this, recall numbers are lower bounds, not true estimates.
High	Remove leiden and gc from default method lists (or add a degenerate-detection guard) — both produce $\text{ARI} = 0$ consistently without tuning.
Medium	Tune Leiden resolution parameter (try 0.05–0.2) on the small run before running large-scale.
Medium	Try more selective blocking (e.g. phonetic key on surname + exact postcode) to improve recall without exploding pair count.
Medium	Run all 5 parties (not just p0 and p1) to assess multi-party linkage performance.
Low	Profile whether the 45-hour large run time is dominated by the failed hclust/pam/gc attempts (expected: yes). Re-time with fixed method list.
Low	Experiment with field-specific similarity (Jaro-Winkler for names, exact match for postcode) rather than auto-detected string similarity.

10 Comparison Across Datasets

NCVR5 at small scale is the strongest result across all three datasets ($\text{ARI} = 0.917$), reflecting a well-structured 4-field linkage task with controlled noise. At large scale the density effect degrades performance substantially. The affiliation dataset remains the hardest challenge ($\text{ARI} = 0.076$), driven by the single-field limitation.

11 Output Files

Metric	CORA	Affiliation	NCVR5 (small / large)
Records	1,879	2,260	10,000 / 100,000
Fields	15	1	4
Task	Dedup	Dedup	Linkage
Best ARI	0.543	0.076	0.917 / 0.195
Best B ³ -F	0.714	0.460	0.963 / 0.715
Best method	SVM	threshold_cc	threshold_cc
Final correct	No (bug)	Yes	Yes / Yes
Main bottleneck	Merge bug, equal weights	Blocking, 1 field	Blocking recall, pair budget

File	Description
small/er_performance.csv	Full metric table — small run (10 methods × 13 metrics)
small/er_predictions.csv	Final cluster assignments — small run (10,000 records)
small/plot_method_comparison.pdf	Grouped bar chart: ARI, B ³ -F, NMI per method (small)
small/plot_performance_heatmap.pdf	Colour heatmap: all metrics × methods (small)
small/plot_cluster_sizes.pdf	Cluster size distribution histogram (small)
small/plot_precision_recall.pdf	Precision vs. Recall scatter (small)
large/er_performance.csv	Full metric table — large run (9 methods × 13 metrics)
large/er_predictions.csv	Final cluster assignments — large run (100,000 records)
large/run_large.log	Full console log of large run
large/plot_method_comparison.pdf	Grouped bar chart (large)
large/plot_performance_heatmap.pdf	Colour heatmap (large)
large/plot_cluster_sizes.pdf	Cluster size distribution histogram (large)
large/plot_precision_recall.pdf	Precision vs. Recall scatter (large)
ncvr5_data_profile.pdf	Dataset profile: field types, distributions, missingness
NCVR5_RESULTS_SUMMARY.pdf	This document